Mirror Descent with Implicit Bias in Deep Learning

DS 608 Paper Presentation

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Presentation Outline

Introduction

Problem Setup and Notation

Stochastic Mirror Descent (SMD)

Implicit Bias: The Linear Case

Nonlinear Analysis: Assumptions and Main Results

Proofs: Bregman Distance Analysis

Experimental Validation

Closing: Summary and Future Work

Question and Answers

Introduction

The Rise of Deep Learning

The 2012 Moment

- AlexNet wins ImageNet 2012 by a 10% margin over second place — the moment the field took notice

What Made It Work?

- Massive overparameterization ($p \gg n$), SGD training, and empirical regularization techniques (dropout, batch norm)

But this raised a puzzle: classical learning theory predicted catastrophic overfitting.

The Generalization Mystery: Zhang et al. (2017)

Same architecture, same optimizer, same hyperparameters. Only the labels changed:

Model	Condition	Train Acc.	Test Acc.
Inception	True labels	100%	∼87%
Inception	Random labels	100%	∼10%
Inception	Random pixels	100%	∼10%
AlexNet	True labels	100%	∼80%
AlexNet	Random labels	100%	∼10%
MLP	True labels	100%	∼53%
MLP	Random labels	100%	∼10%

CIFAR-10. Zhang et al., ICLR 2017.

Model, size, optimizer, regularizer are identical across rows. Only labels differ. None of these explain generalization.

Classical Theory Cannot Explain This

Classical Generalization Bound

$$\text{Test error} \leq \text{Train error} + \underbrace{f(\text{model complexity}, n)}_{\text{complexity penalty}}$$

- Since networks fit *random* labels, their effective VC dimension / Rademacher complexity is essentially unbounded
- Therefore the complexity penalty is vacuously large — the bound predicts the network *should not* generalize even on true labels

The Implication

Networks memorize noise yet generalize on real data. This contradiction vanishes if the optimizer itself creates the bias.

Implicit Bias: The Optimizer as Regularizer

Setup: In the overparameterized regime, the solution set $\mathcal{W}$ is a high-dimensional manifold — infinitely many zero-loss solutions exist.

Known Result (Gunasekar et al., 2018)

For linear models, SGD converges to minimum ℓ_2 norm interpolant by geometry, not explicit constraint.

Three open questions that this paper answers:

1. What is the implicit bias for optimizers *beyond* SGD?
2. Does this extend to *nonlinear* models like deep networks?
3. Can we *deliberately engineer* the implicit bias by choosing the optimizer?

If (3) is yes, the optimizer is a design choice for generalization.

A Unified View: Stochastic Mirror Descent

SGD is just one special case

SMD generalizes all gradient-based optimizers via a potential function ψ

Potential $\psi(w)$	**Algorithm**	**Implicit Bias**
$\frac{1}{2}\|w\|_2^2$	SGD	Min ℓ_2 norm
$\sum_j w_j \log w_j$	Exponentiated Gradient	Sparsity / simplex
$\frac{1}{p}\|w\|_p^p$	p-norm algorithm	Min ℓ_p norm

The choice of ψ is the choice of bias. This paper proves which solution SMD finds for any ψ in nonlinear models.

Problem Setup and Notation

Overparameterized Nonlinear Models

- **Dataset:** $\{(x_i, y_i)\}_{i=1}^n$ where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$.
- **Model:** $f(x_i, w) = f_i(w)$ with parameters $w \in \mathbb{R}^p$.
- **The Interpolating Regime:** $p \gg n$.

The Set of Global Minima (Interpolants)

The set of all parameters consistent with the training data:

$$\mathcal{W} = \{w \in \mathbb{R}^p \mid f_i(w) = y_i, \quad i = 1, \ldots, n\} \tag{1}$$

This forms a high-dimensional $(p - n)$-dimensional manifold.

The Learning Objective

Minimize empirical risk $L(w) = \sum_{i=1}^n L_i(w)$, where $L_i(w) = \ell(y_i, f_i(w))$. Every $w \in \mathcal{W}$ yields $L(w) = 0$.

Optimization via SGD

Standard training typically utilizes **Stochastic Gradient Descent (SGD)**:

$$w_i = w_{i-1} - \eta \nabla L_i(w_{i-1}), \quad i \geq 1 \tag{2}$$

- $L_i(\cdot)$ is differentiable and non-negative.
- $L_i(w) = 0$ iff $y_i = f_i(w)$.
- For $i > n$, we cycle through the data or select at random.

The Implicit Question

Among all $w \in \mathcal{W}$, SGD will converge to exactly one. The update rule's geometry — not any explicit constraint — determines which one. This is the implicit bias.

Stochastic Mirror Descent (SMD)

The Geometry of Mirror Descent

Instead of updating in the weight space, SMD updates in the **mirrored domain** via a strictly convex potential function $\psi(\cdot)$.

The Update Rule

$$\nabla\psi(w_i) = \nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1}) \tag{3}$$

- Updates happen in dual space via $\nabla\psi$
- SGD: $\psi(w) = \frac{1}{2}\|w\|^2$ gives $\nabla\psi = \text{identity}$
- Different ψ encode different geometric biases

Every optimizer is mirror descent in disguise.

The Bregman Perspective

The SMD update can be equivalently viewed as a proximal minimization problem:

$$w_i = \arg\min_{w} \left\{ \eta w^T \nabla L_i(w_{i-1}) + D_\psi(w, w_{i-1}) \right\} \tag{4}$$

Bregman Divergence $D_\psi(w, \bar{w})$

Defined as the difference between the function and its first-order Taylor expansion:

$$D_\psi(w, \bar{w}) := \psi(w) - \psi(\bar{w}) - \nabla\psi(\bar{w})^T (w - \bar{w}) \tag{5}$$

- Non-negative: $D_\psi(w, \bar{w}) \geq 0$, equals 0 iff $w = \bar{w}$
- Tailored to geometry of ψ

Implicit Regularization in Linear Models

Consider the overparameterized linear model $f(x_i, w) = x_i^T w$.

Proposition 1: Convergence to the Closest Interpolant

For a sufficiently small step size η, the SMD iterates converge to:

$$w_\infty = \arg\min_{w \in \mathcal{W}} D_\psi(w, w_0) \tag{6}$$

- Out of infinite $\mathcal{W}$ solutions, SMD picks the Bregman-closest to initialization
- If $w_0 = \arg\min_w \psi(w)$, then $w_\infty = \arg\min_{w \in \mathcal{W}} \psi(w)$
- Geometry is the regularizer

Implication

First rigorous answer to Zhang et al.: optimizer encodes preference. This is determined by ψ and w_0.

Assumptions for Nonlinear Models

Assumption 1: Local Star-Convexity

Setup: Define region $\mathcal{B} = \{w \in \mathbb{R}^p \mid D_\psi(w, \bar{w}) \leq \epsilon\}$ containing w_0.

Assumption Statement: There exists $\bar{w} \in \mathcal{W}$ such that

$$D_{L_i}(\bar{w}, w) := L_i(\bar{w}) - L_i(w) - \nabla L_i(w)^T(\bar{w} - w) \geq 0$$

for all $w \in \mathcal{B}$ and $i = 1, \ldots, n$.

Geometric interpretation: The loss Bregman divergence is non-negative. This means the region near interpolating solutions exhibits local "star-convexity": curvature points inward toward the solution set.

Why it's reasonable: In the overparameterized regime ($p \gg n$), near any solution $\bar{w} \in \mathcal{W}$, the effective loss landscape becomes benign due to the abundance of degrees of freedom.

Assumption 2: Boundedness

Model $f_i(w)$ has bounded derivatives in $\mathcal{B}$.

- Gradient: $\|\nabla f_i(w)\| \leq \gamma$
- Hessian: $\alpha I \preceq H_{f_i}(w) \preceq \beta I$ for constants α, β

These prevent divergence under fixed step size η.

Main Theorems

Theorem 3: Deterministic Convergence

Under Assumption 1, and for a step size η such that $\psi(\cdot) - \eta L_i(\cdot)$ is strictly convex for all i, the following hold for the SMD iterates $\{w_i\}$:

1. **Containment:** All iterates $\{w_i\}$ remain within the region $\mathcal{B}$.
2. **Global Convergence:** The sequence $\{w_i\}$ converges to a limit w_∞.
3. **Interpolation:** The limit point is a global minimum, i.e., $w_\infty \in \mathcal{W}$.

Note: Deterministic convergence, no high-probability bounds needed.

Key point: Bregman geometry guarantees convergence independent of model.

Theorem 4: Characterization of the Limit Point

Let $w^* = \arg\min_{w \in \mathcal{W}} D_\psi(w, w_0)$ be the Bregman-closest interpolant to the initialization.

Proximity to the Optimal Regularized Solution

Under Assumptions 1 and 2, the limit point w_∞ satisfies:

1. $D_\psi(w_\infty, w_0) = D_\psi(w^*, w_0) + o(\epsilon)$
2. $D_\psi(w^*, w_\infty) = o(\epsilon)$

If w_0 is $O(\epsilon)$ from $\mathcal{W}$, then w_∞ is $o(\epsilon)$ from w^*.
Result: SMD implicitly minimizes ψ over interpolants.

Proof Technique: Fundamental Identity

Lemma 6: A General Identity for SMD

The authors derive a fundamental identity that holds for any model $f(\cdot,\cdot)$ and differentiable loss $\ell(\cdot)$. For any $w \in \mathcal{W}$:

$$D_\psi(w, w_{i-1}) = D_\psi(w, w_i) + D_{\psi-\eta L_i}(w_i, w_{i-1}) + \eta L_i(w_i) + \eta D_{L_i}(w, w_{i-1}) \tag{7}$$

Algebraic structure: All three RHS terms are non-negative (by convexity, non-negativity of loss, and star-convexity). This creates a telescoping inequality.

Takeaway: The Bregman distance to any interpolant is non-increasing, with three dissipative mechanisms:

1. Mirror-adjusted divergence shrapes solutions together
2. Loss terms vanish as iterates converge
3. Bregman loss divergence measures local geometry

Lemma 6: Proof Derivation

Start with Bregman definition: $D_\psi(w, w_i) = \psi(w) - \psi(w_i) - \nabla\psi(w_i)^T(w - w_i)$.
Substitute SMD update: $\nabla\psi(w_i) = \nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1})$:

$$D_\psi(w, w_i) = \psi(w) - \psi(w_i) - [\nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1})]^T(w - w_i)$$

Decompose by adding and subtracting $\psi(w_{i-1}) + \nabla\psi(w_{i-1})^T(w - w_{i-1})$:

$$D_\psi(w, w_i) = [D_\psi(w, w_{i-1}) - D_\psi(w_i, w_{i-1})] + \eta\nabla L_i(w_{i-1})^T(w - w_i)$$

Expand $(w - w_i) = (w - w_{i-1}) - (w_i - w_{i-1})$ and apply loss Bregman definitions.
Result: The fundamental identity with telescoping terms.

Lemma 6: Full Mathematical Derivation (Detailed)

Step 1. Expand $D_\psi(w, w_i)$ and $D_\psi(w_i, w_{i-1})$ separately, then combine:

$$D_\psi(w, w_i) = D_\psi(w, w_{i-1}) - D_\psi(w_i, w_{i-1}) + \eta \nabla L_i(w_{i-1})^T (w - w_i) \tag{8}$$

Step 2. For $w \in \mathcal{W}$, use $L_i(w) = 0$:

$$\eta \nabla L_i(w_{i-1})^T (w - w_{i-1}) = \eta L_i(w) - \eta L_i(w_{i-1}) - \eta D_{L_i}(w, w_{i-1})$$

(By definition of Bregman divergence for the loss.)

Step 3. Rearrange using $(w - w_i) = (w - w_{i-1}) - (w_i - w_{i-1})$:

$$\eta \nabla L_i(w_{i-1})^T (w - w_i) = -[L_i(w_{i-1}) - L_i(w_i) - \nabla L_i(w_{i-1})^T (w_i - w_{i-1})] - \eta D_{L_i}(w, w_{i-1})$$

$$= -D_{\psi - \eta L_i}(w_i, w_{i-1}) - \eta L_i(w_i) - \eta D_{L_i}(w, w_{i-1})$$

After rearrangement: the identity follows.

Proof of Theorem 3: Containment and Convergence

Part 1. Containment in $\mathcal{B}$: From Lemma 6 (Fundamental Identity):

$$D_\psi(w, w_{i-1}) = D_\psi(w, w_i) + D_{\psi-\eta L_i}(w_i, w_{i-1}) + \eta L_i(w_i) + \eta D_{L_i}(w, w_{i-1})$$

For $w \in \mathcal{W}$: all RHS terms are non-negative (by strict convexity of $\psi - \eta L_i$, sign of loss, star-convexity assumption). Thus $D_\psi(w, w_i) \leq D_\psi(w, w_{i-1}) \leq \cdots \leq D_\psi(w, w_0) \leq \epsilon$. Iterates stay in $\mathcal{B}$.

Part 2. Convergence: Telescope Lemma 6 over $i = 1, \ldots, T$. Boundary terms cancel:

$$D_\psi(w, w_0) = D_\psi(w, w_T) + \sum_{i=1}^{T} [\text{non-negative terms}]$$

Since $D_\psi(w, w_T) \geq 0$ and LHS is fixed, the sum is bounded: $T \to \infty$ forces each term to 0. In particular, $D_{\psi-\eta L_i}(w_i, w_{i-1}) \to 0$, so $w_i \to w_\infty$ (convergence).

Part 3. Interpolation: Loss terms $\eta L_i(w_i) \to 0$, so $L_i(w_\infty) = 0$ for all i. Thus $w_\infty \in \mathcal{W}$.

Proof of Theorem 4: Closeness to Optimal Point

Setup: Apply Lemma 6 to both w^* and w_∞, both in $\mathcal{W}$. Telescope over all iterations:

$$D_\psi(w^*, w_0) = D_\psi(w^*, w_\infty) + \sum_{i=1}^{\infty}[\cdots]$$

$$D_\psi(w_\infty, w_0) = D_\psi(w_\infty, w_\infty) + \sum_{i=1}^{\infty}[\cdots] = \sum_{i=1}^{\infty}[\cdots]$$

Key observation: The sums depend on w^* and w_∞ only through their distance from intermediate iterates w_{i-1}. By Assumption 2 (bounded Hessian), these distances are $O(\epsilon)$.

Conclusion: Subtracting the two equations and using Hessian bounds via Taylor expansion:

$$D_\psi(w_\infty, w_0) - D_\psi(w^*, w_0) = -D_\psi(w^*, w_\infty) + o(\epsilon)$$

Since both $D_\psi(w_\infty, w_0) - D_\psi(w^*, w_0) \geq 0$ (by definition of w^*) and $D_\psi(w^*, w_\infty) \geq 0$, each is $o(\epsilon)$.

Highly Overparameterized Regime: Why w_0 Is Close to $\mathcal{W}$

In the limit $p \gg n$, a random initialization is whp close to the interpolating set.

Linear Approximation Argument

- Distance to $\mathcal{W}$ is approximated by the solution to the linearized problem:

$$\min_{w} \|w - w_0\|^2 \quad \text{s.t.} \quad y \approx f(x, w_0) + \nabla f(x, w_0)^T (w - w_0) \tag{9}$$

- The Jacobian $\nabla f(x, w_0)$ is $n \times p$, consisting of sums of outer products
- When $p \gg n$ and data are sufficiently random: $\lambda_{\min}(\nabla f^T \nabla f) = \Omega(p)$
- Thus: $\|w^* - w_0\|^2 \approx \|y - f(x, w_0)\|^2 \cdot O(1/p) = O(n/p)$
- Conclusion: w_0 is $O(\sqrt{n/p})$ away from the manifold—negligible as p grows

Experimental Validation

Experimental Setup

Datasets: MNIST (60K samples), CIFAR-10 (50K samples). Both highly overparameterized.

Architectures: MNIST: 4-layer CNN with 2M parameters. CIFAR-10: ResNet-18 with 11M parameters.

Protocol: Random initialization $\mathcal{N}(0, 0.0001)$, train to 100% accuracy.

Seeds: 6 initializations (MNIST), 8 initializations (CIFAR-10).

Algorithms: Mirror potentials $\psi(w) = \frac{1}{q}\|w\|_q^q$ for $q \in \{1+\epsilon, 2, 3, 10\}$.

Experimental Hypothesis

The paper predicts that **each SMD algorithm finds the interpolant closest to its initialization in the corresponding Bregman divergence**.

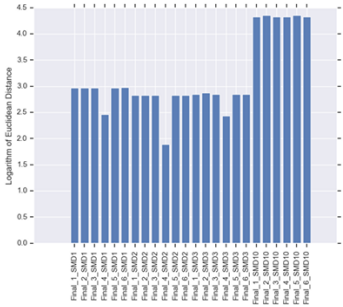
Testing Strategy

For each initialization w_0 and each mirror type (with potential ψ):

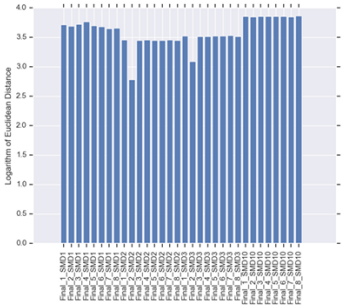
1. Train the model to zero training error, reaching some $w_\infty \in \mathcal{W}$
2. Compute Bregman divergences $D_{\psi_q}(w_\infty^{(q)}, w_0)$ for all mirrors $q \in \{1+\epsilon, 2, 3, 10\}$
3. Check: Is the minimum always on the diagonal?

- **Prediction:** For fixed w_0 and fixed ψ_q, the closest final point to w_0 (in D_{ψ_q} distance) should be the one obtained by running SMD with ψ_q
- This would confirm both convergence to w^* and the theoretical characterization

MNIST Results: Convergence to Nearest Interpolant



MNIST. SGD Starting from Initial 4

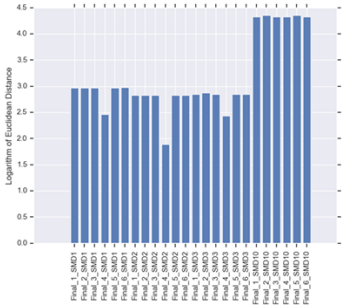


CIFAR-10. SGD Starting from Initial 2

Figure 5: Distances between a particular initial point and all the final points obtained by both different initializations and different mirrors. The smallest distance, among all initializations and all mirrors, corresponds exactly to the final point obtained from that initial point by SGD. This trend is observed consistently for all other mirror descents and all initializations (see the results in Tables 8 and 9 in the appendix).

Diagonal dominance: Each mirror finds its closest interpolant. Confirms Theorem 4.

CIFAR-10 Results: Consistency at Scale



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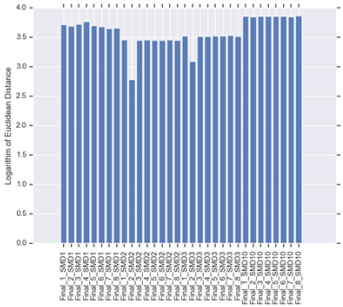

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Diagonal pattern holds at scale: ResNet-18, 11M parameters, 8 seeds. Compatible with

Mirror Effects on Final Weight Distributions

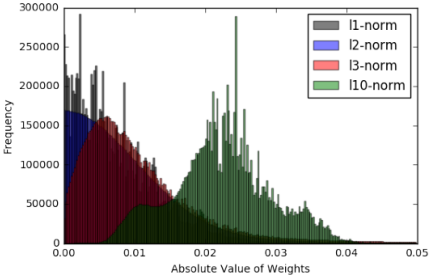

Figure 6: Histogram of the absolute value of the final weights in the network for different SMD algorithm with different potentials. Note that each of the four histograms corresponds to an 11×10^6-dimensional weight vector that perfectly interpolates the data. Even though the weights remain quite small, the histograms are drastically different. ℓ_1-SMD induces sparsity on the weights, as expected. SGD does not seem to change the distribution of the weights significantly. ℓ_3-SMD starts to reduce the sparsity, and ℓ_{10} shifts the distribution of the weights significantly, so much so that almost all the weights are non-zero.

Generalization Performance Across Mirrors

MNIST: All mirrors converge to $\geq 99\%$ test accuracy. Implicit bias does not hurt MNIST generalization.

CIFAR-10: Mirror choice significantly impacts test error:

- ℓ_{10}-SMD: Best generalization (densest weights, fewest constraints)
- ℓ_2 (SGD): Intermediate performance
- ℓ_3-SMD: Slightly worse than ℓ_2
- ℓ_1-SMD: Worst (extreme sparsity overly regularizes)

Insight: Implicit bias controls generalization. Denser regularization regimes generalize better on complex tasks. Optimizer is a design lever for test performance.

Summary of Contributions

Summary

- **Implicit Bias**: Exact characterization in linear case, approximate in nonlinear
- **Star-Convexity**: Local regularity connects linear theory to deep learning
- **Optimizer as Regularizer**: Choice of ψ determines bias and generalization
- **Unified View**: SGD, exponentiated gradient, p-norm algorithms all fit framework

Generalization is the geometric footprint of the optimizer.

Future Work

1. Adaptive optimizers: What is effective ψ for Adam or RMSProp?
2. Transformer architectures: Does analysis extend to non-Euclidean geometry?
3. Tighter bounds: Can $o(\epsilon)$ be exact for specific architectures?
4. Learning ψ: Adapt mirror map during training?

Question and Answers